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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MTH 213 — Past Questions & Materials
by Renance

Course: MTH 213
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 6
Levels: 200

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Paper 1 of 6 — MTH 213 Linear Algebra 1 First Semester 2018 2019

Level 200 • First Semester • PastQuestion

; ; ; and assume t

solution for $x \in \mathbb{R}$ if $t \neq 0$. Show that the equation ax

7

line L with equation $2x -$

vector space.

(C) Let $V = \mathbb{R}^2$ show that the

$\{0\}$ consists of all vectors of the form

$(t; 2t)$ with $t \in \mathbb{R}$,

Question 2 (4) marks

(a) Let $S = \{w, \dots, b\}$ in V . (73/19)

1) Define a sequence in V . When is S said to be linearly (i) dependent, (ii) independent?

(b) Determine whether

the following three vectors in $\mathbb{R}^3$ are linearly independent or not?

$\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$

$\begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 6 \end{pmatrix}$ in $\mathbb{R}^3$ — Thanks

3/5 marks

*, determine the intersection $U \cap V$ of the subspaces

$U = \{x + 2y = 0\}$ in $\mathbb{R}^3$. $V = \{w, x, y, z\} \in \mathbb{R}^4 : wt + x + 2y = 0\}$ — Thanks

(a) Define the coordinate of a vector space V relative to its basis B — CGMKs

(b) For which values of a does the system have a unique solution, and for which pairs of values (a, b)

$x + ay = 73$

does each system in unknowns x and y have more than one solution? f

$2x + 3y = b$ for all b

(c) Consider the basis $B = \{b_1, b_2\}$ for $\mathbb{R}^2$, where $b_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$, $b_2 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$; Suppose that $x \in \mathbb{R}^2$ has B -coordinate vector $[x]_B = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$. Find x

& $\forall y \in \mathbb{R}$

Question 4 (1 mark)

(a) Define the Nullity of a Matrix A — 4 marks

$\begin{pmatrix} -3 & 6 & -11 & -7 \end{pmatrix}$

(b) Find the nullity of the matrix $\begin{pmatrix} 1 & -2 & 2 & 3 \end{pmatrix}$ — 3 marks

$\begin{pmatrix} 2 & -4 & 5 & 8 & -4 \end{pmatrix}$ is

(c) Prove that if V be a vector space. If W, U, W , & V are subspaces of V , then $W \cap (U + W)$ is a subspace of V . — 3 marks

Question 5

(a) Define a linear transformation, — Om/K«

Consider the linear transformation $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $f(x, y)$,
vector spaces $\mathbb{R}^2$ and $\mathbb{R}^2$.

(b) Write down the matrix of f with respect to the standard bases of $\mathbb{R}^2$ and $\mathbb{R}^2$, — mks

(c) Determine the matrix $T[f]S$, where S and T are the bases $S: (1, 0, 1), (0, 1, 1), (0, 0, 1)$, $F: (2, 2), (2, 2)$. — 0

$Z) = (2x - 3y + 7z \ x - Z)$ between the real

Question 6 |

(a) Define the Kernel of a linear transformation. — 5 foams

| Oy, 2 . 7

(b) Taking $= \mathbb{R}$, find the row and column ranks of the matrix $A = f |$ — FY¥omics

RS |

ned by the vectors

(c) Consider the real vector space $\mathbb{R}^3$ and let W' be the subspace spanned by

the vectors — IO BmKg:

1d (ii) dimension for W' ?

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Paper 2 of 6 — MTH213 2017-2018

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN.
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS AND COMPUTER SCIENCE
FIRST SEMESTER EXAMINATION 2017/2018 SESSION
COURSE TITLE & CODE: LINEAR ALGEBRA I (MTH 213)
Instruction: Attempt any four questions. Time: 2 hours

Question One

- a. If the inverse of matrix A is $A^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{14} & -\frac{5}{14} \\ -\frac{1}{2} & \frac{3}{14} & -\frac{1}{14} \\ \frac{1}{2} & -\frac{1}{14} & \frac{5}{14} \end{bmatrix}$, find the matrix A. (Hint: $A \cdot A^{-1} = I_3$)
- b. Verify if the set of vectors $S = \{((-1, 3, 4), (2, 5, -3), (4, 1, 6))\}$ in $\mathbb{R}^3$ is linearly dependent or independent

Question Two

- a. Reduce the matrix $B = \begin{bmatrix} 4 & 8 & 2 & 3 \\ 19 & 41 & 1 & -2 \\ -1 & 2 & -1 & -3 \\ 2 & 13 & 1 & 2 \end{bmatrix}$ to a row-reduced echelon form
- b. Determine if the vector $w = (40, 21, -17)$ is a linear combination of $u_1 = (1, 1, -3)$, $u_2 = (2, 1, 3)$, $u_3 = (10, 5, -2)$

Question Three

- a. Given the matrix $B = \begin{bmatrix} 4 & 8 & 2 & 3 \\ 19 & 41 & 1 & -2 \\ -1 & 2 & -1 & -3 \\ 2 & 13 & 1 & 2 \end{bmatrix}$, find the determinant of B.
- b. Show that the set of vectors $A = \{(2, 1, 2), (3, 3, 1), (4, -1, 0)\}$ forms a basis for $\mathbb{R}^3$

Question Four

- a. Let $X = \mathbb{R}^2$ and $A = \begin{bmatrix} 1 & 4 \\ 2 & 8 \end{bmatrix}$. Define $T: X \rightarrow X$ by $Tz = Az$, where $z = \begin{bmatrix} x \\ y \end{bmatrix} \in X$. Show that T is a linear transformation and hence find the Kernel and Range of T
- b. Let $A = (2, 3, 4)$ and $B = (8, -2, 5)$ be two vectors, find the cross (vector) product of A and B

Question Five

- a. Find (i) AB and (ii) BA if it exists of the partitioned matrices $A = \begin{bmatrix} 3 & 0 & -1 & 2 & 1 \\ 1 & 5 & -3 & 4 & 2 \\ -1 & -5 & 2 & 7 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & -1 & 3 \\ -2 & 2 & 2 \\ 1 & -3 & 0 \\ -3 & 6 & 7 \\ -1 & 1 & 4 \end{bmatrix}$
- b. (i) Prove or disprove that the sets of Real numbers $\mathbb{R}$ and Natural numbers $\mathbb{N}$ are vector spaces
(ii) State any five properties of determinants and justify it with an example each.

Paper 3 of 6 — MTH 213 2022-2024

Level 200 • First Semester • PastQuestion

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DEPARTMENT OF MATHEMATICS
FIRST SEMESTER EXAMINATION 2022/2023 SESSION
COURSE TITLE & CODE: LINEAR ALGEBRA I (MTH 213)

Instruction: Attempt any four questions Time: 2 hours

Question 1 (17.5 marks)

- (a) Let $X = \mathbb{R}^2$ and $A = \begin{bmatrix} 2 & 5 \\ -2 & 3 \end{bmatrix}$. Define $T: X \rightarrow X$ by $Tz = Az$, where $z = \begin{bmatrix} x \\ y \end{bmatrix} \in X$. Show that T is a linear transformation and hence find the Kernel and Range of T (12.5 marks)
- (b) Let $A = (2, 3, 4)$ and $B = (8, -2, 5)$ be two vectors, find the cross (vector) product of A and B ($A \times B$) (5 marks)

Question 2 (17.5 marks)

Given the following matrices $A = \begin{bmatrix} 2 & -2 \\ 3 & 1 \\ -1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} -4 & 6 & -5 \\ 5 & -3 & 2 \end{bmatrix}$, $C = \begin{bmatrix} -3 & -2 \\ 4 & 6 \\ 8 & -2 \end{bmatrix}$, $D = \begin{bmatrix} -2 & -2 & 4 \\ 6 & 1 & 3 \end{bmatrix}$. Evaluate

- (a) (i) AB (ii) $(BA)^2$ (9 marks)
- (b) (i) $(DC)^2$ (ii) CD (8.5 marks)

Question 3 (17.5 marks)

- (a) Find the determinant of the matrix $A = \begin{bmatrix} 3 & 5 & 7 & 2 \\ 2 & 4 & 6 & 1 \\ 1 & 0 & 2 & 4 \\ 5 & 3 & 1 & 2 \end{bmatrix}$ (10 marks)

- (b) List 5 properties of determinants and justify each property with an example. (7.5 marks)

Question 4 (17.5 marks)

- (a) Find the inverse of A (A^{-1}) given that $A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 5 & 3 \\ -1 & 1 & 2 \end{bmatrix}$ (12.5 marks)

- (b) Given $B = \begin{bmatrix} -4 & 6 & -5 \\ 5 & -3 & 2 \end{bmatrix}$, $C = \begin{bmatrix} -3 & -2 \\ 4 & 6 \\ 8 & -2 \end{bmatrix}$, find the trace of the transpose of the matrix BC . (5 marks)

Question 5 (17.5 marks)

- (a) Define the following subsets of Real numbers

(i) Natural numbers (ii) Whole numbers (iii) Integers (iv) Rational numbers (4 marks)

- (b) Write the vector W as a linearly combination of the vectors V_1, V_2 and V_3 . (13.5 marks)

$$V_1 = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}, V_2 = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix}, V_3 = \begin{pmatrix} -4 \\ 3 \\ 4 \end{pmatrix} \text{ and } W = \begin{pmatrix} 9 \\ 0 \\ -2 \end{pmatrix}.$$

Question 6 (17.5 marks)

- (a) Define a vector space and list all its axioms.

- (b) Determine if the vectors V_1, V_2 and V_3 are linearly independent, where

$$V_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, V_2 = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}, V_3 = \begin{pmatrix} -3 \\ 1 \\ 1 \end{pmatrix}$$

(6 marks)

(11.5 marks)

Paper 4 of 6 — MTH 213 PQ 2024

Level 200 • First Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
COLLEGE OF SCIENCE
DEPARTMENT OF MATHEMATICS
FIRST SEMESTER EXAMINATION 2023/2024 SESSION
COURSE TITLE & CODE: LINEAR ALGEBRA I (MTH 213)

Instruction: Attempt any four questions Time: 2 hours

Question 1

Given the following matrices $A = \begin{bmatrix} 2 & -2 \\ 3 & 1 \\ -1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} -4 & 6 & -5 \\ 5 & -3 & 2 \end{bmatrix}$, $C = \begin{bmatrix} -3 & -2 \\ 4 & 6 \\ 0 & -2 \end{bmatrix}$, $D = \begin{bmatrix} -2 & -2 & 4 \\ 6 & 1 & 3 \end{bmatrix}$

Evaluate

(a) $A + (CBA)^T - CB(DC)^T$ (9 marks)

(b) Reduce the matrix $B = \begin{bmatrix} 3 & 5 & 7 & 2 \\ 2 & -4 & 6 & 1 \\ 1 & -1 & 2 & 4 \\ 5 & 3 & 1 & -2 \end{bmatrix}$ to its echelon and hence find its rank. (8.5 marks)

Question 2

(a) Find the determinant of the matrix $B = \begin{bmatrix} 3 & 5 & 7 & 2 \\ 2 & -4 & 6 & 1 \\ 1 & -1 & 2 & 4 \\ 5 & 3 & 1 & -2 \end{bmatrix}$ (10 marks)

(b) List 5 properties of determinants and justify each property with an example. (7.5 marks)

Question 3

(a) Find the inverse of A (A^{-1}) given that $A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 4 & 5 \\ 2 & 3 & -6 \end{bmatrix}$ (12.5 marks)

(b) Given $B = \begin{bmatrix} -4 & 6 & -5 \\ 5 & -3 & 2 \end{bmatrix}$, $C = \begin{bmatrix} -3 & -2 \\ 4 & 6 \\ 0 & -2 \end{bmatrix}$, find the trace of the transpose of the matrix BC . (5 marks)

Question 4

(a) Let $X = \mathbb{R}^2$ and $A = \begin{bmatrix} 2 & 5 \\ -2 & 3 \end{bmatrix}$. Define $T: X \rightarrow X$ by $Tx = Ax$, where $x = \begin{bmatrix} x \\ y \end{bmatrix} \in X$. Show that T is a linear transformation and hence find the Kernel and Range of T (12.5 marks)

(b) Let $A = (2, 3, 4)$ and $B = (6, -2, 5)$ be two vectors, find the cross (vector) product of A and B . (5 marks)

Question 5

(a) Define the following (i) Natural numbers (ii) Subspace (iii) Integers (iv) Rational numbers (4 marks)

(b) Write the vector W as a linearly combination of the vectors V_1, V_2 and V_3 . (13.5 marks)

$$V_1 = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}, V_2 = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix}, V_3 = \begin{pmatrix} -4 \\ 3 \\ 4 \end{pmatrix} \text{ and } W = \begin{pmatrix} 9 \\ 0 \\ -2 \end{pmatrix}.$$

Question 6

(a) Define a vector space and list all its axioms. (6 marks)

(b) Determine if the vectors V_1, V_2 and V_3 are linearly independent, where (11.5 marks)

$$V_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, V_2 = \begin{pmatrix} 2 \\ -2 \\ 1 \end{pmatrix}, V_3 = \begin{pmatrix} -3 \\ 4 \\ 1 \end{pmatrix}$$

Paper 5 of 6 — MTH 213

Level 200 • First Semester • Handout

CHAPTER

M AT R I X

X_ REPRESENTATION OF LINEAR

TRANSFORMATIONS

I INTRODUCTION

we shall

In this chapter, study the relationship

between linear transformation

rices.

In particular,

we shall show that every linear an

transformation

could

represented by a matrix and vice versa. This will be achieved by establishing a one-to-

one correspondence between the set of linear transformations and the set of $n \times n$ matrices over the field F .

matrices. $M_n(F)$ over the field F .

Theorem 5. 1

Suppose X be a finite dimensional vector space over the field

Suppose

F and suppose X

is an ordered basis for X . Let Y be a vector

space over the same field, F and

let y be any given vector in Y . Then there exists a unique linear transformation

$T: X \rightarrow Y$ such that

$Tx_j = y_j, j = 1, 2, 3, \dots, n$.

Proof

Suppose $x \in X$ be arbitrary. Then there exists a unique n -tuple a such that

$x = \sum_{j=1}^n a_j x_j$,

Define $T: X \rightarrow Y$ by

$Tx = \sum_{j=1}^n a_j y_j$

$x \in X$. then $Tx = \sum_{j=1}^n a_j y_j$, showing that T is well defined and clearly, $Tx_j = y_j$. Next

show that T is linear. For this, let $x, z \in X$ and $a, b \in F$ (field) be arbitrary.

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1

Chapter : Matrix Representation of Linear Transformations

$Tx =$

$a_1 y_1 +$

$a_{12}t + a_{1t} +$

$a_{1Y'm}$

$.Tx =$

$12Y t$

$a_{2y2} + a_{32Yat} + a_{m2Y m}$

$Tx a_{13y} + a_{a3\%} + d_{3y's t} + a_{13}$

$Tx,, =$

$ay_1 ta_{2nY2} + a_{31Y3 t} \dots^*$

$pmYm$

In other words,

$1x, = 2ayy; a_{1j)y} + a_{2\%2 t} + a_{gy3 t}$

$+ a_{uy Ym}^{**}$

Thus, it implies that a_j , a , $Gays \dots$, i are the coordinates of $7x$, in the ordered basis P_y and the scalars a could easily be obtained from equation (5.1) to form the $m \times n$ matrix

$a_{g a_{23} a_{2n}}$

$3n$

$m \times n$

Now, let $r \in X$ be arbitrary. Then x

$a, + a_{\%2} tay + \dots + C$, where

$K =$

is the coordinate matrix of x . Hence

This implies that

158 Linear Algebra

AK 24,

Jl

is the

coordinate matrix of Tr in the ordered matrix

β_y . Thus, the $m \times n$

equation (5.1) determine the linear transformation T . Now, the matrix A scalars a , in is called the

MATRIX OF T relative to the

pair of the ordered bases α_r and β_y .

5.3 UNIQUENESS OF THE REPRESENTATION

Let $B =$

(b_i) be another matrix representation of T (discussed

ordered bases B_x above) relative to the

we obtain AK

and β_y . If $x \in X$ is arbitrary with

coordinate matrix K ,

$BK =$

Now, if $X =$

$X + X, + X, t \dots + X$

with the coordinate vector

we get $a_j =$

b_j , and so $A =$

B . Thus, the matrix

representation is unique.

5.4 REPRESENTATION OF A MATRIX BY A LINEAR

On the other hand,

if $A(a_j)$ is $m \times n$ matrix, TRANSFORMATION

then define $T: X \rightarrow Y$ by

$T(x) =$

(where $x = x_j \in X$ is arbitrary)

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4-(-),

$a_{11} \dots a_{1n}$

Example 5.4

1)((-

be ordered basis for R and

respectively. Find the matrix A which represents the linear transformation

$T: R \rightarrow R$ defined by

2-

$= |x+2$

$3y$

Solution

Let $u = 2$

and $v = \{y\} = \{$

$w,$

$Tu = \sum_{j=1}^n a_{kj} v_j$

$k =$

Then

$Tu = a_1 + a_2 + a_3 + \dots$

-3

$a_0 + a_1 + a_2 + a_3 + \dots$

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(ii)

(iii)

(6) Find the matrix representation of T if the same bases in (a) above are used but $7T$ becomes

A13

7x

$a+2, x+2x^2+ +34,$

$tX^2+X^3|$

Chapter 5: Matrix Representation of Linear Transformations

4-141[

a)

2o

B

1LT[21

0

(6)

2

--JU-|lo

3f41[21-

3

(c) 2

3

13. Suppose a linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$

T is defined by

$T(x) = 3x$

(a) Find the matrix representation of T if the basis for $\mathbb{R}^n$ and $\mathbb{R}^m$ are respectively

$\{1, 0, 0\}$

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1 Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation defined by

$T(x) = 2x + x^2$

$4x - 3$

$+2x$

Find the matrix representation of T if the basis for $\mathbb{R}^n$ and $\mathbb{R}^m$ are respectively

$\{1, 2, -1\}$ and $\{1, 0, 2, 3\}$

0

(a)

1fo][5

1

(b) $3 - 1$.

221) 4

12. Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear transformation defined by

$T(x, y, z) = (2x, y, 3z)$

$T(x, y, z) = (2x, y, 3z)$

Chapter 5: Matrix Representation of Linear Transformations

2-1] o]

a) (b)

4] [0] [5]

(c) (d)

Resolve No 8 above if T is redefined by

9.

$X - y - z = 32$

(i) $2x + 3y + 6z$ (i)

$-4x + 2$

10 Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear transformation defined by

$T(x, y, z) = (x + 3y, x - y + 2z, 3x + 2y + 4z)$

$5x - y + 2$

$3x + 2y + 4z$

involved are

of T if the basis

Find the matrix representation

Or

(a)

1 for $\mathbb{R}$ and $\mathbb{O}$

34

for

For $\mathbb{R}'$ and $|\mathbb{R}'|$

(6) -11

$31 - \frac{2}{3} || 4$

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. matrix

representation of $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$

' , a linear transformation defined

by

$T(x, y, z) = (3x - 5y, 2x + 3y, 8x - 3y)$ relative to the basis

e. . where $e_1 =$

$(1, 0)$. $e_2 =$

$(0, 1)$ for $\mathbb{R}^3$ and

$a = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ for P .

Resolve No 3 above with basis $e_1 = (1, 2)$, $e_2 = (-1, 3)$ for R^* and $\{(5, 1, 3)$.

4.

$\{(-1, 2, 1), (0, 2, -3)\}$ for R'' .

5. Let $T: R \rightarrow R'$ be a linear transformation. Find the matrix representation of T

if T is defined by $T(x, y, z) = (x + 3y, 2x - 4y + z, 6x - y + 4z)$ and

basis for R is $\{(1, 1, 1), (2, 1, 0), (0, 3, 1)\}$.

6. Let $X = R$ and $Y = R^*$, Define $T: X \rightarrow Y$ by $T(x, y) = (x + y, x + y, x)$. Let

$B = \{(1, 2), (-1, 3)\}$ and $C = \{(1, 1, 0), (0, 1, 1), (1, 0, 1)\}$. Find the matrix

representation of T using

(a) B as a basis for R and C for R'

(b) the standard basis for X and Y

(c) the standard basis for X and C for R'

(d) B for the basis of X and the standard basis for Y .

7. Let $T: R \rightarrow R'$ be defined by

$T(x, y) = (x - y, 3x + 2y)$ and $T: R \rightarrow R'$ by $T(x, y) = (-5y, 3x + y)$

(a) Find the matrix representation A_1 and A_2 of T and T , respectively with respect to the

(i) standard basis of

R (ii) basis $\{(1, 4), (6, 3)\}$ for R .

(b) Determine the matrix representation of (i) T , (ii) T^{-1} ;

(ii) T^{-1} , with respect to the standard basis

of R .

d. Let $T: R \rightarrow R'$ be a linear transformation defined by

$T(x, y, z) = (x + y + z,$

$7y, x^2 - y^2)$

$y + 2z$:

$T: R \rightarrow R'$ by $T(x, y, z) = (-3x + 3y - z,$

Find the matrix of T with respect to the following basis for P' .

Transformations 13

Chapter 5: Matrix Representation of Linear

$2a^4 + 2a^4 + a^4$

a^{14}

E .

$ta^{34} + a$

$344t^2a^4 + 4a^4 + 2a^4$

$-2434 + 34t^2a$

$a^{14}t^2 + 3a^{24} + 20s^4$

This implies that

20142024 tau-0

$$-3a + az^4 + 4a + 20 = -3$$

$$-2a^2 + 3az \tau au = 12$$

$$a^4 + 30,4$$

$$2a^4 = 14$$

Solving simultaneously gives

607

159 355 256 as 43

a1s 86 24 86 ,au 43

matrix is

Summarily, the required

550 539

503

40 43 86

153 351 355|

40 86 86

$A = () - 13 \ 185 \ 256$

43 43 43

607

821 1072 32

43 43

20 43

EXERCISES5 $(-1, 0), (1, 2, 1)\}$ be a basis for R and

$(0, 1), (2,$

Let $a, x_2, x_i\} = \{(1,$

the linear

basis for R ". Find

$\{(2, 1, 0), (1, 2, 0), (1,$

$-1, 3)\}$ be a

Yes 1, 2, 3.

j

$T: R^n$ such R that $7x, y \gg =$

Transformation

that $\{1, 1, 1\}$

$(2, 3).$

transformation such

$2 R$: is a linear find 7.

if $T: R^n$ and $7(-2, 1, 0)$

$(6, 1).$

$(1, 0, 1) = (4, -1)$

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+a

$2012 + 2a^2$

13

$-3a^{12} + a^2 + 4a + 20 =$

-7

$-2a^2 + 3a + a$

-1

20

$a^1 + 3a$

Solving simultaneously gives

T13 1072

550

$a^{12} 43 424 43 a^{42} 43$

Next, we have

$Tu 73 2j=k3 =$

$as t asV2 t a3s"3 t a43a$

|20

| 2 0 + 2 +a

$-3a^{13} t az + 4a + 2a$

+a²³

$-202+ 3a^{33} t 043$

$ais t302 -2as3$

This implies that

$2a^{13} + 2ay + a^{33} 25$

$-3a + ay + 40 + 204 = 1$

$-2023 + 3a^3y t aa^3 = 4$

$as + 3a^3 - 2a 20$

Solving simultaneously results in

-S39 351 185

86 a³³ =

4 3 = 32

43

Finally, we have

0

$Tu, 7 12 2 4 a^{14} + az^4V2 + a^3a^3 +$

0,s'4

-2

hapler 5: Malrix Representation of Linear Tanstormalions

1057

A

l-17]

2 t 2a2 t a

-30 t a2+ 4a t 20,41

-2a +30u +41

a +3d21-2a1

by coniparison, that

his implies

2a+2a +ai = 23

t a t 403 +20,, 9

-30

-2 +3a +

aa = 4

1 +30-2a,, = -17

Solving simultaneously gives

503 49

d 40 20

Furthermore, we havee

1 13 = aj2 +ayV2 + a32V'3 + a42V4

Tuy=7

-2

2412 + 2a + d3

-3a2 + a2+ 402 203

t

4 2

12 + a22 +a32*ds2 2

t

3d

202

d

S implies that

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0 0

"0 u o

Comparing we get

a = -2

a 4 6 , a4=1, 0u ="0,

Compiling a,s, we obtain the required matrix

13 3 0 6

4 0 -1 1

$A = \begin{pmatrix} 1 & 3 & 0 & 6 \\ 4 & 0 & -1 & 1 \end{pmatrix}$

0

0 50 -2

Example 5.6

Find the matrix representation of the linear mapping of Example 5.5 above in the basis for $\mathbb{R}^n$.

Solution

Let the basis be

$\{ \begin{bmatrix} 2 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} \}$

$\{ \begin{bmatrix} 4 \\ -3 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} \}$

Recall that

$Tu_j = \sum_{k=1}^n a_{kj} u_k$

$k=1$

1, 2,

3, 4

Then for $x_j =$

$\begin{bmatrix} 2 \\ 1 \\ 1 \\ 0 \end{bmatrix}$

$\begin{bmatrix} -3 \\ X \\ 0 \\ 1 \end{bmatrix}$

we have

Chapter 5: Matrix Representation of Linear Transformations 165

$Tu_j = \sum_{k=1}^n a_{kj} u_k$

$Tu_j =$

$\begin{bmatrix} 0 \\ 2 \\ a_2 \\ 0 \end{bmatrix}$

a_2

0

Thus, by

comparison we obtain

$a_{23}, a_{03}, a_{23} =$

3, $a_{g3} =$

5

Next, we have

0

$Tu_j =$

$\begin{bmatrix} a_{23} \\ a_{13} \\ a_{23} \\ a_{33} \end{bmatrix}$

$\begin{bmatrix} a_{13} \\ a_{23} \\ a_{33} \end{bmatrix}$

0

before, by comparison. we obtain

$=0$

$a_{13} = 0, a_{21} = -1, a_{33} = 6, a_{34} = 0$

Finally, we get.

$a_{11} = 0$

$T_{ij}, k=1$

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$4x^3 + 3x^2 + 6x,$

$4x^2$

$x^3 + 3x^2 + 6x^3$

$s, 5x - 2x$

Solution

Let the standard basis for R be

$\{e_1, e_2, e_3\}$

Using equation (5.1) we obtain

$T_{ij}, i, j = 1, 2, 3, 4$

Thus.

13

0 4

This implies that $a_{11} = 13, d_{14} = 1, a_{14} = 0$

Again, we obtain

b

Chapter 5. Matrix Representation of Linear Transformations

comparison that

implies by

is a 3×3 matrix.

$a - 2a_1 =$

-1

-3

solving simultaneously gives

Next, we have

L_0 .

$= a_1 + 2$

$0 \text{ or } -a_{12}$

This implies by comparison, that

$a_{12} = a_2 - 2$

$G_2 - 2a_1 = -1$

$a_{12} = 0$

Solving simultaneously we obtain

$a_{10} = -3, a_2 = -1$

CDmple the a,'s to get the required nmatrix

3 01

A()1-3

0

Crample 5.5

heordered basis for R' find the maurix .representation of the linear
astormation T:R R and defined by

Transformattons

Chapler 5: Matrix Representabion of Linear

. , y. ' (0, 0, 0), (0. 1,0; 0). (0, 0., l. 0), (0. 0, 0, 1)} respectively.

2.j= 1,2,3

=1

=0,2 0, we have

Eor ,y

- T(1,0,0):= (1,0, 2,-)=24 = adv, t az2 t agi¥s t G41Va

10, 0. 0)+ a2(0.1,0, 0)+4a,(0.0,1, 0) + ai(0, 0,0,1)

-(1.a a1,as= RHS

has. comparing (1. 0. 2. -1) with RHS we have

a 1, a20, d 2, a= -1

Next we obtain for x =0, y

1,2 =

0

Ti, T(0,1,0):= (-2.1.1,0) Ea

k=

an1,0, 0,0) a,(0, 1,0,0) a 0,0, 1,0) a(0,0,0, 1)

RHS

(2 22 ap.g2)

oby comparison of (-2. 1. 1. 0) with RHS we have

a2-2, d2

1, a2= 1, a4 =0

urtherinore, we get for x = 0. y = 0, 7 *l,

T(0,0, 1):= (1.3,- 4, 1) = a%3" =a t a t a3sVs t a,a'a .

aue(l,0,0,0)+ aza(0. 1, 0,0) +as(0, 0, 1. 0) + a0,0, 0, 1)

(;3.d23,3.4s3)= RHS

nparison, of (1, 3, 4, 1) with RHS we obtain

OTE, the

s 1 ,4n 3, dau -4, a = =

l

required matrix becomes

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Thus,

$$T_h = T(\cdot, X_3)$$

$$(2x, +3r, +3s) \cdot (0, 1, 2, 0) + (-x, +,) \cdot (6, 0, 0, 2)$$

$$+ (24, 2, +, -1, 1, 4, -3)$$

$$(0, x, +x + \cdot)^* \cdot d + fx, +tiy \cdot (0) - (-31, 34, \cdot 0, 0, -4 \cdot 1)$$

Example 5.2

Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear transformation such that $T(1, 0, 0) = (2, 1)$,

$T(0, 1, 0) = (0, 3)$, and $T(0, 0, 1) = (4, 6)$. Find T .

Solution

Recall that

the set $\{x, d_2, X_3\} = \{(1, 0, 0), (0, 1, 0),$

$(0, 0, 1)\}$ is a basis for $\mathbb{R}^3$.

If $v =$

$x, y, z \in \mathbb{R}^3$ is arbitrary, then there exist scalars $a, b,$

$$v = (x, y, z)$$

such that

$$v = a(1, 0, 0) + b(0, 1, 0) + c(0, 0, 1) \quad (a, b, c \in \mathbb{R})$$

This implies that

$x = a, b = y$ and $c = z$. So that

$$Tv =$$

$$T(x, y, z) = aT(1, 0, 0) + bT(0, 1, 0) + cT(0, 0, 1)$$

$$= x(2, 1) + y(0, 3) + z(4, 6) = (2x + 4z, x + 3y + 6z)$$

Example 5.3

Suppose $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is defined by

$$T(x, y, z) = (r - 2y + z, y + 3z, 2x + y - 4z, -x + z).$$

Determine the matrix representation of T with

basis of $\mathbb{R}^3$ and $\mathbb{R}^3$ respectively, respect to the standard ordered

Solution

Let the standard basis for $\mathbb{R}^3$ and be

$$\mathcal{B} = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$$

Chapter 5: Matrix Representation of Linear Transformations

linear.

Then, T is

The following theorem establishes our discussion so far.

Theorem 5.1

Let X and Y be vector spaces over a field F with dimensions n and m

Suppose

respectively. If $\mathcal{P}_X$ and $\mathcal{P}_Y$ are the ordered bases for X and Y respectively; then for each

NEAR TRANSFORMATION, $T: X \gg Y$. there exists a UNIQUE $m \times n$ MATRIX A

LINE

in entries in F such that, for arbitrary $x \in X$ with coordinate matrix $B = [s_i]$, the

with

coordinate matrix $[7 \times 1]_g$, is given by

Conversely, let $M_n(F)$ be the set of all $n \times n$ matrices over the field F and

Let $L(X, Y)$ be the set of all linear transformations from X to Y , then the mapping

$g(LAX, Y)$. M, F defined by $g(7) A =$

is an isomorphism where $TE L\{X, Y\}$ is arbitrary and A is the matrix representation of T .

Example

L,

5. I^*

$X_2, X_3\} = \{(0, 1, 1), (+1, 0, 2), (2, 1, 1)\}$ be a basis for R and let

in R. Find the

$s = \{(0, 1, 2, 0), (6, 0, 0, 2), (-1, 1, 4, -3)\}$ be vector

a

linear transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ such that $T(x_j) = x_{j+1}$ for $j = 1, 2, 3$.

Solution

Let R' be arbitrary. Then

there exist scalars a, d, a , such that

Let $\mathbf{x} = (x_1, \dots, x_n)$

$$X = x_3) = a_1(0, 1, 1) + a_2(-1, 0, 2) + a_3(2, 1, 1)$$

(-2 +20, +0, C, +202 +03)

This implies

$$-4 + 20X$$

Solving ultaneously for 2, gives

$$3(2x, +5x, t23), a, \{(-*+*3)$$

and d

$$(2x - x^2 + tA^3)$$

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where $x =$

a, x , and $z > B, x$. Then,

$$T(\text{exe}) = 7a_{\text{us}} + 2B_{B,,}$$

jl

$$\sum_{j=1}^n j =$$
$$a_{\alpha} + B_{\alpha} T_7 = 2(a_{\alpha} + B_{\alpha}) y_{\alpha} = 2 a_{\alpha} y_{\alpha} + 2 B_{\alpha} y_{\alpha}$$
$$j=1 \quad j=1$$
$$0a + PZBy, = a + BT:]$$

Thus, T is linear. Finally, we show that T is unique. For this, assume that there exists another linear transformation $Q: T \rightarrow Y$ such that $Qx = y$. Let $x = 20x$, $\rightarrow X$ be an arbitrary vector. Then
Or
 $\forall j=1, \dots, n$, $-ay, Tx$. Thus, $Q = T$ and hence T is unique. This completes the proof.

5.2 MATRIX REPRESENTATION OF A

Suppose X and Y are vector spaces over $\mathbb{F}$ with TRANSFORMATION

Suppose also that B_X and B_Y are ordered bases for X and Y respectively. If $T: X \rightarrow Y$ is linear

a transformation, then $Tx = \sum_{j=1}^n a_{ij} b_j$, $i=1, 2, 3, \dots, n$.
 $R(T)Y$

So, each Tx , could be expressed as linear combination
a

Hence there exist y_i , for Each.
scalars $\alpha_i = \sum_{j=1}^n a_{ij}$, $i=1, 2, 3, \dots, n$

such that

Paper 6 of 6 — MTH 213 Note

Level 200 • First Semester • Handout

CHAPTER

M AT R I X

X_ REPRESENTATION OF LINEAR TRANSFORMATIONS

I INTRODUCTION

we shail

Inthis chapter. study the relationship
between linear transformation
rices.

In particular,

we shall show that every linear an
transformation

could

nrresented by a matrir and vice versa. This will be achieved by establishing a oneDe-
e correspondence between the set of linear transformations and the set ot m KU
matrices. MP) over the field. F.

Theorem 5. 1

ppose Xbe a finite dimensional vector space over the field

Suppose

F and supposeX

is an ordered basis tor x. Let Y be a vector
space over the same field, F and

kt be any given vector in Y. Then there exists a unique linear uransformation
T:X Y such that

$Tx_j = y_j, j = 1, 2, 3, \dots, n.$

Proos

Suppose $x \in X$ be arbitrary. Then there exists a unique n-tuplea such that

a,

Define $T:X \rightarrow Y$ by

$j=1, \dots, n$

X . then $Tx_j = y_j$,. showing that T is wel defined and clearly, $Ty_j = y_j$,. Next
W that T is linear. For this, let $x, z \in X$ and $a, b \in F$ (field) be arbitrary.

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1

nppler : Matrix Representation of Linear Transformations

$Tx =$

$a_1 y +$

$a_2 y^2 + a_1 t +$

$a_m y^m$

$.Tx =$

$12Y t$

$a_2 y^2 + a_3 y^3 + a_2 y^m$

$Tx = a_1 y + a_2 y^2 + a_3 y^3 + \dots + a_n y^n$

$Tx, =$

$a_1 y + a_2 y^2 + a_3 y^3 + \dots$

$p_m y^m$

In other words,

$1x, = 2a_1 y + a_2 y^2 + a_3 y^3 + \dots$

$+ a_n y^n$

Thus, it implies that $a_1, a_2, \dots, a_n$ are the coordinates of Tx , in the ordered basis β_y and the scalars a_i could easily be obtained from equation (5.1) to form the $m \times n$ matrix

$A = [a_{ij}]$

$n \times n$

$m \times n$

Now, let $x \in X$ be arbitrary. Then $x =$

$a_1 y + a_2 y^2 + \dots + a_n y^n$, where

$K =$

K is the coordinate matrix of x . Hence

This implies that

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AK 24,

J

is the

coordinate matrix of Tx in the ordered matrix

β_y . Thus, the $m \times n$

equation (5.1) determine the linear transformation T . Now, the matrix A of scalars a_i , in is called the

MATRIX OF T relative to the

pair of the ordered bases β_x and β_y .

5.3 UNIQUENESS OF THE REPRESENTATION

Let $B =$

(b_i) be another matrix representation of T (discussed

ordered bases β_x above) relative to the

we obtain AK

and βy . If $x \in X$ is arbitrary with
coordinate matrix K ,
 $BK =$

Now, if $X =$
 $x_1 + x_2 + \dots + x_n$
with the coordinate vector
we get $a_j =$
 b_j , and so $A =$
 B . Thus, the matrix
representation is unique.

5.4 REPRESENTATION OF A MATRIX BY A LINEAR

On the other hand,
if $A = (a_{ij})$ is $m \times n$ matrix, TRANSFORMATION
then define $T: X \rightarrow Y$ by
 $T(x) = Ax$,
(where $x \in X$ is arbitrary)

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$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$,
 $A^{-1} = \begin{pmatrix} 1 & -2 \\ -3 & 1 \end{pmatrix}$

Example 5.4

$\{1, x, x^2\}$ be ordered basis for R and
respectively. Find the matrix A which represents the linear transformation
 $T: R \rightarrow R$ defined by

$T(x^2) = x^2 + 2x$
 $T(x) = 3x$

Solution

Let $u = x^2$ and $v = x$, $w = 1$
 $T(u) = u + 2v = x^2 + 2x$
 $T(v) = 3v = 3x$
 $T(w) = w = 1$

Then
 $T(u) = 1 \cdot u + 2 \cdot v + 0 \cdot w$
 $T(v) = 0 \cdot u + 3 \cdot v + 0 \cdot w$
 $T(w) = 0 \cdot u + 0 \cdot v + 1 \cdot w$

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(ii)

(iii)

(6) Find the matrix representation of T if the same bases in (a) above are used but $7T$ becomes

A_{13}

$7x$

$a+2, x+2x_2+ +34,$

$tX_2+X_3|$

Chapter 5: Matrix Representation of Linear Transformations

4-141[

a)

2o

B

$1LT[21$

0

(6)

2

$-JU-|lo$

$3f41[21-$

3

(c) 2

3

13. Suppose a linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$

R is defined by

$Lar+3$

(a) Find the matrix representation of T if the basis for $\mathbb{R}^n$ and $\mathbb{R}^m$ are respectively

olo

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1 Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation defined by

$2x + xs$

$4x-3$

$+2x$

Find the matrix representation of T if the basis for $\mathbb{R}^n$ and $\mathbb{R}^m$ are respectively

$o [21-2] -1|02||3||$

0

(a)

$1fo][5$

1

(b) 3 -1.

221) 4

12. Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear transformation defined by

$T(x, y, z) = (x + 2y, x - 2y, x - 2z)$

$T(x, y, z) = (x + 2y, x - 2y, x - 2z)$

Chapter 5: Matrix Representation of Linear Transformations

2-1] o]

a) (b)

4] [0] [s5]]

(c) (d)

Resolve No 8 above if T is redefined by

9.

$T(x, y, z) = (x - y - z, x + 2y, x - 2z)$

(i) $2x + 3y + 6z$ (i)

$-4x + 2$

10 Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear transformation defined by

$T(x, y, z) = (x + 3y, x - y + 2z, 3x + 2y + 4z)$

$5x - y + 2$

$3x + 2y + 4z$

involved are

of T if the basis

Find the matrix representation

Or

(a)

1 for $\mathbb{R}$ and 0

34

for

For $\mathbb{R}'$ and $|\mathbb{R}'|$

(6) -11

$31 - \frac{2}{3} || 4$

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. matrix

representation of $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$

' , a linear transformation defined

by

$T(x, y, z) = (3x - 5y, 2x + 3y, 8x - 3y)$ relative to the basis

e_1, e_2, e_3 where $e_1 =$

$(1, 0, 0)$. $e_2 =$

$(0, 1)$ for R^* and

$a = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ for P .

Resolve No 3 above with basis $e, = (1, 2), e, = (-1, 3)$ for R^* and $\{(5, 1, 3),$

4.

$-1, 2, 1), (0, 2, -3)\}$ for R'' .

5. Let $T: R^3 \rightarrow R^3$ be a linear transformation. Find the matrix representation of T

if T is defined by $T(x, y, z) = (x + 3y, 2x - 4y + z, 5x - y + 4z)$ and

basis for R is $\{(1, 1, 1), (2, 1, 0), (0, 3, 1)\}$.

6. Let $X = R$ and $Y = R^*$, Define $T: X \rightarrow Y$ by $T(y) = (r + y, r + i, y)$. Let

$B = \{1, 2\}$, $C = \{(1, 1, 0), (0, 1, 1), (1, 0, 1)\}$. Find the matrix

representation of T using

(a) B as a basis for R and C for R^*

(b) the standard basis for X and Y

(c) the standard basis for X and C for R^*

(d) B for the basis of X and the standard basis for Y .

7. Let $T: R^3 \rightarrow R^3$ be defined by

$T(x, y, z) = (x - y, 3x + 2y, 5x - 3y + z)$ and $T: R^3 \rightarrow R^3$ by $T(x, y, z) = (-5y, 3x + z)$

(a) Find the matrix representation A , and A , of T and T , respectively with respect to the

(i) standard basis of

R (ii) basis $\{(1, 4), (6, 3)\}$ for R .

(b) Determine the matrix representation of (i) T , + T (ii) $T - T$;

(ii) $T - T$, with respect to the standard basis

of R .

d. Let $T: R^3 \rightarrow R^3$ be a linear transformation defined by

$T(x, y, z) = x + y + z$

$T(y, z) = 2x -$

$y + 2z$:

$T(x, y, z) = -3x + 3y - z$

Find the matrix of T with respect to the following basis for P^3 .

Transformations 13

Chapter 5: Matrix Representation of Linear

$2a^4 + 2a^4 + a^4$

a^{14}

$E.$

$ta^{34} + a$

$344t^{aa} + 4a + 2a$

$-2434 + 34ta$

$a^{14}t^{3a}2420s^4$

This implies that

20142024 tau-0

$-3a + az^4 + 4a + 20 = -3$

$-2a^2 + 3az^t au = 12$

$a^4 + 30,4$

$2a^4 = 14$

Solving simultaneously gives

607

159 355 256 as 43

als 86 24 86 ,au 43

matrix is

Summarily, the required

550 539

503

40 43 86

153 351 355|

40 86 86

$A = () - 13 \ 185 \ 256$

43 43 43

607

821 1072 32

43 43

20 43

EXERCISES5 $(-1, 0), (1, 2, 1)\}$ be a basis for R and

$0, 1), (2,$

Let $a, x_2, x_i\} = \{(1,$

the linear

basis for R ". Find

$\{(2, 1, 0), (1, 2, 0), (1,$

$-1, 3)\}$ be a

Ys 1, 2, 3.

j

$T: R^n$ such R that $7x, y \gg =$

Transformation

that $\{1, 1, 1\}$

$(2, 3).$

transformation such

2 R : is a linear find 7.

it $T: R^n$ and $7(-2, 1, 0)$

$(6, 1).$

$(1, 0, 1) = (4, -1)$

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+a

$2012 + 2a^2$

13

$-3a^{12} + a^2 + 4a + 20 =$

-7

$-2a^2 + 3a + a$

-1

20

$a^1 + 3a$

Solving simultaneously gives

T13 1072

550

$a^{12} 43 424 43 a^{42} 43$

Next, we have

$Tu 73 2j=k3 =$

$as t asV2 t a3s"3 t a43a$

|20

| 2 0 + 2 +a

$-3a^{13} t az + 4a + 2a$

+a²³

$-202+ 3a^{33} t 043$

$ais t302 -2as3$

This implies that

$2a^{13} + 2ay + a^{33} 25$

$-3a + ay + 40 + 204 = 1$

$-2023 + 3a^3y t aa^3 = 4$

$as + 3a^3 - 2a 20$

Solving simultaneously results in

-S39 351 185

86 a³³ =

4 3 = 32

43

Finally, we have

0

$Tu, 7 12 2 4 a^{14} + az^4V2 + a^3a^3 +$

0,s'4

-2

hapler 5: Malrix Representation of Linear Tanstormalions

1057

A

l-17]

$2t + 2a + t + a$

$-30t + a^2 + 4a + t + 20, 41$

$-2a + 30u + 41$

$a + 3d + 21 - 2a + 1$

by comparison, that

this implies

$2a + 2a + a + i = 23$

$t + a + t + 403 + 20, 9$

-30

$-2 + 3a +$

$aa = 4$

$1 + 30 - 2a, = -17$

Solving simultaneously gives

503 49

d 40 20

Furthermore, we have

$1 + 13 = a_j^2 + a_y V^2 + a_{32} V'^3 + a_{42} V^4$

$T_{uy} = 7$

-2

$2412 + 2a + d^3$

$-3a^2 + a^2 + 402 + 203$

t

4 2

$12 + a^2 + a_{32} ds^2 + 2$

t

3d

202

d

S implies that

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0 0

"0 u o

Comparing we get

$a = -2$

$a + 4 + 6, a_4 = 1, 0u = "0,$

Compiling a,s, we obtain the required matrix

13 3 0 6

4 0 -1 1

$A = \begin{pmatrix} 1 & 3 & 0 & 6 \\ 4 & 0 & -1 & 1 \end{pmatrix}$

0

0 50 -2

Example 5.6

Find the matrix representation of the linear mapping of Example 5.5 above in the basis for $\mathbb{R}^n$.

Solution

Let the basis be

$\{ \begin{bmatrix} 2 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} \}$

$\{ \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \\ 0 \end{bmatrix} \}$

Recall that

$Tu_j = \sum_{k=1}^n a_{jk} u_k$

$k=1$

1, 2,

3, 4

Then for $x = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$,

$2. X = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$

$-3, X, 0, \dots = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$

1, we have

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$22 = a_1 x^2 + a_2 x + a_3 x^2 + a_4 x^3$

$Tu = 0$

0 2

a_2

0

Thus, by

comparison we obtain

$a_2 = 3, a_0 = 0, a_2 = 5$

3, $a_3 = 5$

5

Next, we have

0

Tu

$[a_2 \ 3 \ 5 \ 0]$

$a_1 \ 2 \ a_2 \ 3 \ a_3 \ 4 \ a_4 \ 3 \ a_5 \ 3$

0 10

before, by comparison. we obtain

$=0$

$a_{13} = 0, a_{21} = -1, a_{33} = 6, a_{34} = 0$

Finally, we get.

$a_{11} = 0$

$T_{ij}, k=1$

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$4x^3 + 3x^2 + 6x,$

$4x^2$

$X^3 + 3x^2 + 6x^3$

$s, 5x - 2x$

Solution

Let the standard basis for R be

$\{e_1, e_2, e_3\}$

Using equation (5.1) we obtain

$T_{ij}, i, j = 1, 2, 3, 4$

Thus.

13

$0, 4$

This implies that $a_{11} = 13, d_{14} = 1, a_{14} = 0$

Again, we obtain

b

Chapter 5. Matrix Representation of Linear Transformations

comparison that

implies by

is that $a_{11} = 3$.

$a_{11} - 2a_{11} =$

-1

-3

solving simultaneously gives

Next, we have

L_0 .

$= a_{11} + 2$

$0, a_{12}$

This implies by comparison, that

$a_{12} = a_{21} - 2$

$G_2 - 2a_{11} = -1$

$a_{12} = 0$

Solving simultaneously we obtain

$a_{10} = -3, a_{21} = -1$

CDmple the a,'s to get the required nmatrix

3 01

A()1-3

0

Crample 5.5

heordered basis for R' find the maurix .representation of the linear
astormation T:R R and defined by

Transformattons

Chapler 5: Matrix Representabion of Linear

, , y. ' (,0, 0, 0), (0. 1,0; 0). (0, 0., l. 0), (0. 0, 0, 1)} respectively.

2.j= 1,2,3

=1

=0,2 0, we have

Eor ,y

- T(1,0,0):= (1,0, 2,-)=24 = adv, t az2 t agi¥s t G41Va

10, 0. 0)+ a2(0.1,0, 0)+4a,(0.0,1, 0) + ai(0, 0,0,1)

-(1.a a1,as= RHS

has. comparing (1. 0. 2. -1) with RHS we have

a 1, a20, d 2, a= -1

Next we obtain for x =0, y

1,2 =

0

Ti, T(0,1,0):= (-2.1.1,0) Ea

k=

an1,0, 0,0) a,(0, 1,0,0) a 0,0, 1,0) a(0,0,0, 1)

RHS

(2 22 ap.g2)

oby comparison of (-2. 1. 1. 0) with RHS we have

a2-2, d2

1, a2= 1, a4 =0

urtherinore, we get for x = 0. y = 0, 7 *l,

T(0,0, 1):= (1.3,- 4, 1) = a%3" =a t a t a3sVs t a,a'a .

aue(l,0,0,0)+ aza(0. 1, 0,0) +as(0, 0, 1. 0) + a0,0, 0, 1)

(;3.d23,3.4s3)= RHS

nparison, of (1, 3, 4, 1) with RHS we obtain

OTE, the

s 1 ,4n 3, dau -4, a = =

l

required matrix becomes

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Thus,

$$T_h = T(\cdot, X_3)$$

$$(2x, +3r, +3s) \cdot (0, 1, 2, 0) + (-x, +,) \cdot (6, 0, 0, 2)$$

$$+ (24, 2, +, -1, 1, 4, -3)$$

$$(0, x, +x + \cdot)^* \cdot d + fx, +tiy \cdot (0) - (-31, 34, \cdot 0, 0, -4 \cdot 1)$$

Example 5.2

Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear transformation such that $T(1, 0, 0) = (2, 1)$,

$T(0, 1, 0) = (0, 3)$, and $T(0, 0, 1) = (4, 6)$. Find T .

Solution

Recall that

the set $\{x, d_2, X_3\} = \{(1, 0, 0), (0, 1, 0),$

$(0, 0, 1)\}$ is a basis for $\mathbb{R}^3$.

If $v =$

$x, y, z \in \mathbb{R}^3$ is arbitrary, then there exist scalars a, b, c ,

$$v = (x, y, z)$$

such that

$$v = a(1, 0, 0) + b(0, 1, 0) + c(0, 0, 1) \quad (a, b, c \in \mathbb{R})$$

This implies that

$x = a, y = b, z = c$ So that

$$Tv =$$

$$T(x, y, z) = aT(1, 0, 0) + bT(0, 1, 0) + cT(0, 0, 1)$$

$$= a(2, 1) + b(0, 3) + c(4, 6) = (2x + 4z, x + 3y + 6z)$$

Example 5.3

Suppose $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is defined by

$$T(x, y, z) = (r - 2y + z, y + 3z, 2x + y - 4z, -x + z).$$

Determine the matrix representation of T with

basis of $\mathbb{R}^3$ and $\mathbb{R}^3$ respectively, respect to the standard ordered

Solution

Let the standard basis for $\mathbb{R}^3$ and be

$$\mathcal{B}_1 = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$$

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linear.

Then, T is

The following theorem establishes our discussion so far.

Theorem 5.1

Let X and Y be vector spaces over a field F with dimensions n and m

Suppose

respectively. If $\mathcal{P}_X$ and $\mathcal{P}_Y$ are the ordered bases for X and Y respectively; then for each

NEAR TRANSFORMATION, $T: X \rightarrow Y$. there exists a UNIQUE $m \times n$ MATRIX A

LINE

in entries in F such that. for arbitrary $x \in X$ with coordinate matrix $B = [s]$. the with

coordinate matrix $[x]$, is given by

Conversely. let $M_n(F)$ be the set of all $m \times n$ matrices over the field F and

$L(X, Y)$ be the set of all linear transformations from X to Y , then the mapping

$g: L(X, Y) \rightarrow M_n(F)$ defined by $g(T) = A$

is an isomorphism where $T \in L(X, Y)$ is arbitrary and A is the matrix representation of T .

Example

L ,

5. I^*

$\{x_1, x_2, x_3\} = \{(0, 1, 1), (1, 0, 2), (2, 1, 1)\}$ be a basis for R and let

in R . Find the

$s = \{(0, 1, 2, 0), (6, 0, 0, 2), (-1, 1, 4, -3)\}$ be vector

a

linear transformation $T: R \rightarrow R^*$ such that $T(x_j) = \sum_{i=1}^4 s_{ij}$.

Solution

Let R' be arbitrary. Then

there exist scalars a, b, c, d , such that

Let $x = (x_1, x_2, x_3, x_4)$

$x = x_3 = a(0, 1, 1) + b(-1, 0, 2) + c(2, 1, 1)$

$(-2, 20, 0, 0) = a(0, 1, 1) + b(-1, 0, 2) + c(2, 1, 1)$

This implies

$-4 + 20x$

Solving simultaneously for 2 , gives

$3(2x, +5x, 23), a, \{(-1, +1, 3)\}$

and d

$(2x - x_2, 2a_3)$

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where $x =$

a, x , and $z \in B, x, \dots$. Then,

$T(xe) = 7a_{us}, +2B_{B,,}$

$j|$

$\forall j=1, j=$

$a, +B, T, = 2(a_{\ll}, +B_{B,,})y, = 2aa y + 2B, y,$

$j=| j=|$

$oa + PZBy, = a + BT: j$

Thus, T is linear. Finally, we show that T is unique. For this, assume that there exists another linear transformation $Q: T \rightarrow Y$ such that $Qx = y$. Let $x = 20x$, $\rightarrow X$ be an arbitrary vector. Then
Or
 $\forall j=1, \dots, n$, $-ay, Tx$. Thus, $Q = T$ and hence T is unique. This completes the proof.

5.2 MATRIX REPRESENTATION OF A

Suppose X and Y are vector spaces over $\mathbb{F}$ with TRANSFORMATION

Suppose also that B_X and B_Y are ordered bases for X and Y respectively. If $T: X \rightarrow Y$ is linear

a transformation, then $Tx = \sum_{j=1}^n a_{ij} b_j$, $i=1, 2, 3, \dots, n$.
 $R(T)Y$

So, each Tx , could be expressed as linear combination
a

Hence there exist y_i , for Each.
scalars $\alpha_i = \sum_{j=1}^n a_{ij}$, $i=1, 2, 3, \dots, n$

such that